

Exercise Problems

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Chapter 1

Notes and Introduction

This Lab course is intended as an introduction to computational methods useful for solving physical problems. *C* is the preferred programming language, and *Linux* the preferred OS. A knowledge of both is assumed. Those not familiar with either of the above are expected to learn it by themselves.

The Lab programme consists of a sequence of problems, divided into three chapters, which are to be implemented/solved by computer programming. Some are intended to help you learn a few subtleties about the way computers work, while others are problems you should be familiar with from Physics. On completion of problems in each chapter, a report of each one is to be prepared using $\text{\LaTeX}$ and submitted in PDF format ($\text{\LaTeX}$ will **not** be taught either). The report must contain, apart from the problem, the code you have written, result obtained, and a discussion on the result leading to what you have gathered from the exercise. Course evaluation will be based on the lab report, and a final examination.

Chapter 2

Basics of numerical computing

Algorithms of a few problems are given below, often with sketchy details of what it tries to do. Implement them using *C*, and analyze your results.

2.1 Problem 1: Precision in your machine

Encode the following algorithm and run it to determine the smallest positive number that can be represented on the computer you are using.

```
input  $s < - - -1.0$ 
for  $k = 1, 2, 3, \dots, 100$  do
     $s < - - -0.5s$ 
     $t < - - -s + 1.0$ 
    if  $t \leq 1.0$  then
         $s < - - -2.0s$ 
        output  $k - 1, s$ 
        stop
    endif
end
```

Do this for both single precision and double precision floating point numbers.

2.2 Problem 2

Evaluate the expression $y = (x^2 + 1.0)^{0.5} - 1.0$ in two ways

$$\begin{aligned} \text{(a.) } y &< -(x^2 + 1.0)^{0.5} - 1.0 \\ \text{(b.) } y &< -\frac{x^2}{(x^2 + 1.0)^{0.5} + 1.0} \end{aligned} \tag{2.1}$$

for small values of x , $x = 0.1$, 0.01 and 0.001 . Determine the fractional error in both the methods of performing the subtraction.

Which method is superior, and why?

2.3 Problem 3

It is desired to calculate all integral powers of the number

$$x = (\sqrt{5} - 1)/2.$$

It turns out that the integral powers of x satisfy a simple recursive relation

$$x^{n+1} = x^{n-1} - x^n$$

Show that the above recurrence relation is unstable by calculating x^{16} , x^{30} , x^{40} and x^{50} from the recurrence relation and comparing with the actual values.

2.4 Problem 4

Given $x_n = (1/3)^n$, it turns out that the integral powers of $(1/3)$ satisfy a simple recursive relation $x_{n+1} = (13/3)x_n - (4/3)x_{n-1}$ with $x_0 = 1$ and $x_1 = (1/3)$. Show that the above recurrence relation is unstable by calculating x_{15} , x_{30} , x_{40} and x_{50} from the recurrence relation and comparing with the actual values.

2.5 Problem 5

The recurrence relation

$$y_{n+1} = e - (n+1)y_n$$

(where e is the base of natural logarithm) can be obtained from integration by parts to the integral

$$y_n = \int_0^1 x^n e^x dx. \tag{2.2}$$

Show that the above recurrence relation is unstable by calculating y_{15} and y_{20} from the recurrence relation.

2.6 Problem 6

Compute the dot product of the following two vectors

$$x = [2.718281823, -3.141592654, 1.414213562, 0.5772156649, 0.3010299957] \quad (2.3)$$

and

$$y = [1486.2497, 878366.9879, -22.37492, 4773714.647, 0.000185049] \quad (2.4)$$

Compute the summation in four ways

1. forward order Summation $\{x_i y_i\}$ for $i = 1, ..n$
2. reverse order Summation $\{x_i y_i\}$ for $i = n, ..1$
3. largest to smallest order (add positive numbers in order from largest to smallest, then add negative numbers in order from smallest to largest and then add the two partial sums)
4. smallest to largest order (reverse order of adding in the previous method)

Use both single and double precision for a total of eight answers. Compare the results with the correct value $1.006571E - 9$.

Chapter 3

Projectile Motion: Solving differential equations

Refer to Sections 2.3-2.5 of the lecture notes on Computational Physics for the theory, and definition of parameters. The integration has to be done using Euler method (see Section 1.1 of the lecture notes).

3.1 Problem 1: Free projectile

Solve the two dimensional trajectory of a projectile fired from a cannon shell located at the origin using the Euler method. Assume the initial projected speed is 700 meter/s , and using different firing angles starting from 20 degrees to 60 degrees with an interval of 5 degrees. Neglect the effects of the air resistance. Plot the trajectories of the projectile for different firing angles. Also plot the range of the projectile against the firing angles and show numerically that the maximum range of the projectile corresponds to a firing angle of 45 degrees. Compare the numerical results with the analytical solutions as far as the range and the duration of the projectile.

3.2 Problem 2: Air resistance

Extend the above problem to include the effect of the air resistance by assuming a value of $B/m = 4 * 10^{-5} \text{ meter}^{-1}$. Plot the trajectories for a given firing angle for the case of no air resistance and with air resistance. Repeat this for different firing angles. Also plot the range of the projectile against the different firing angles and find numerically the firing angle which corresponds to the maximum range of the projectile.

3.3 Problem 3: Inhomogeneous air resistance

Now consider the air resistance is not uniform. Consider the air density varies as $\exp(-y/y_0)$, where $y_0 = 10^4 \text{ meters}$. Now plot the trajectories for a given firing angle without air resistance,

with constant air resistance and with varying air resistance as given above. Repeat this for different firing angles. Also plot the range of the projectile against the different firing angles and find numerically the firing angle which corresponds to the maximum range of the projectile.

Chapter 4

Integration

4.1 Problem1: Celestial Motion

Describe the motion of a planet for each of the following sets of initial condition. The sun is assumed to be at the origin and its motion is not considered. The only force involved being the force of gravitation. Assuming $Gm_1 = 1$, where G is gravitation constant & m_1 is the mass of sun, one can avoid dealing with astronomical magnitudes for positions:

(i) $x_0 = 0.5$; $y_0 = 0.0$; $v_{0,x} = 0.0$; $v_{0,y} = 1.63$

(ii) $x_0 = 0.5$; $y_0 = 0.0$; $v_{0,x} = 0.0$; $v_{0,y} = 1.80$

(iii) $x_0 = 0.5$; $y_0 = 0.0$; $v_{0,x} = 0.0$; $v_{0,y} = 1.40$

(iv) $x_0 = 0.5$; $y_0 = 0.0$; $v_{0,x} = 0.50$; $v_{0,y} = 1.80$

(v) $x_0 = 0.5$; $y_0 = 0.0$; $v_{0,x} = -0.50$; $v_{0,y} = 1.40$

(vi) $x_0 = 0.5$; $y_0 = 0.50$; $v_{0,x} = -0.50$; $v_{0,y} = 0.50$

4.2 Problem 2: Trapezoidal & Simpson's rule

Write two functions

```
float trap(float (* func)(float), float a, float b, int N)
```

```
float simp(float (* func)(float), float a, float b, int N)
```

which can be used to integrate the function `float func(float)` from the lower limit `a` to upper limit `b` using `N` intervals and using the Trapezoidal and Simpson rule respectively.

4.3 Problem 3: Applications

Use the functions written in problem 4.2. to evaluate the following integrals.

f(x)	x	x^2	sin(x)	x sin(x)
a	0	0	0	0
b	1	1	π	π

Evaluate each integral using both the trap() and simp() using $N = 2, 4, 8, \dots, 1024$. Let y_{NUM} be the numerical result. All these integrals can be done analytically, so let y_{AN} be the analytic result. We define the error as

$$E(N) = (y_{NUM} - y_{AN})/y_{AN}$$

where the error depends on N , the number of intervals used. Show plot of $E(N)$ with varying N for all the integrals. Are these results in keeping with how you expect the error to scale with N ?

4.4 Problem 4: Integrand with a singularity

Consider the integral

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx \quad (4.1)$$

1. How will you evaluate this integral analytically?
2. How will you evaluate this integral numerically?
3. Write a program to evaluate this integral numerically? Compare your result with the analytical result.

Chapter 5

Some advanced problems

5.1 Bound motion

We consider the 1-dimensional motion of a particle of mass m in a time independent potential $V(x)$. The fact that the energy E will be conserved allows us to integrate the equation of motion and obtain a solution in a closed form

$$t - C = \sqrt{\frac{m}{2}} \int_a^{-a} \frac{1}{\sqrt{E - V}} dx \quad (5.1)$$

where for the choice $C = 0$ the particle is at the position x_i at the time $t = 0$ and x refers to its position at any arbitrary time t . We consider a particular case where the particle is in bound motion between two points a and b where $V(a) = E$ and $V(b) = E$ and $V(x) \leq E$ for $a \leq x \leq b$. The time period of the oscillation T is given by

$$T = 2\sqrt{\frac{m}{2}} \int_a^{-a} \frac{1}{\sqrt{E - V}} dx \quad (5.2)$$

First consider a particle with $m = 1$ Kg in the potential $V(x) = (1/2)\alpha x^2$ with $\alpha = 4$ Kgs². Numerically calculate the time period of oscillation and check this against the expected value. Verify that the frequency does not depend on the amplitude of oscillation. Next consider a potential $V(x) = \exp((1/2)\alpha^2 x^2)$. Numerically verify that for small amplitude oscillations you recover the same results as the simple harmonic oscillator. The *floatpot(float)*; time period is expected to be different for large amplitude oscillations. How does the time period vary with the amplitude of oscillations? Show this graphically.

5.2 Trajectory in an arbitrary potential

Write a program using Eulers Method and using the 2nd Order Runge-Kutta Method to find the phase space trajectory of a particle of unit mass in a 1-D potential $V(x) = (x^2)/2$. a. Show the trajectory for the initial conditions $(x, p) = (1, 0.1), (-1, 0.1), (1, 10)$. b. Check the conservation of energy for different step sizes, for both the Euler Method and the 2nd Order Runge Kutta Method. Determine the step size where you have at least 1

5.3 Trajectory of a charged particle

Write a program to follow the motion of an electron in an electric field $E(x, t)$ and a magnetic field $B(x, t)$. Numerically determine the trajectory of an electron which starts at the origin with velocity $v = (1, 1, 1)$ m/sec for the following field configurations a. Uniform magnetic field 10^{-4} Tesla along the z axis. b. Uniform magnetic field 10^{-6} Tesla along the z axis and an uniform electric field 1V/m along the Y axis.

5.4 Random walk

A drunkard takes a random walk from the origin, randomly moving one step left or right. Using the `rand()` function in C, simulate the motion of a one dimensional random walker.

- i) Find the average, and root mean square distance traveled by the random walker in time t .
- ii) Repeat the same as i) in 2-dimensions.